

Physics for Engineers 2 Lab

VOLTAGE VARIATIONS IN CAPACITORS

MPS Department | FEU Institute of Technology



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OBJECTIVES

- a. To define time constant RC
- b. To determine charging and discharging voltage of a capacitor
- c. To plot the voltage against time during charging and discharging



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Time Constant

The time required to charge a capacitor to about 63% of the maximum voltage in an RC circuit is called the time constant of the circuit. It is the product of Resistance and Capacitance

$$\tau = RC$$

Where

τ is the time constant (s)

R is the resistance (Ω)

C is the capacitance (F)

It takes 5τ for a capacitor to reach a steady state or full charge value.



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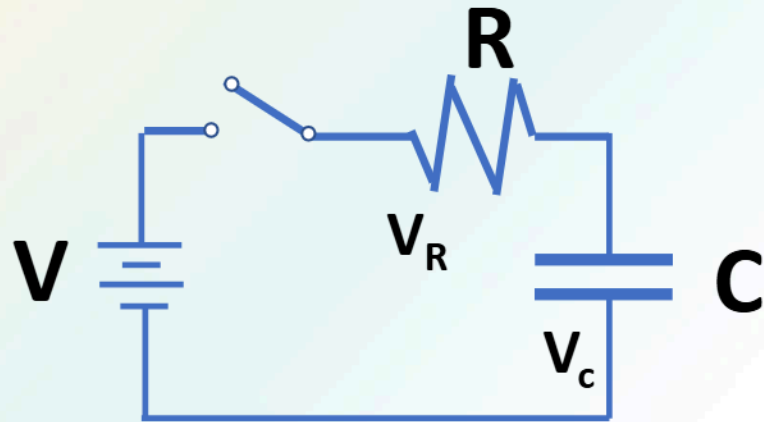


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Charging Capacitors

When a capacitor charges, electrons flow onto one plate and move off the other plate. This process will be continued until the potential difference across the capacitor V_c is equal to the potential difference across the battery, V .

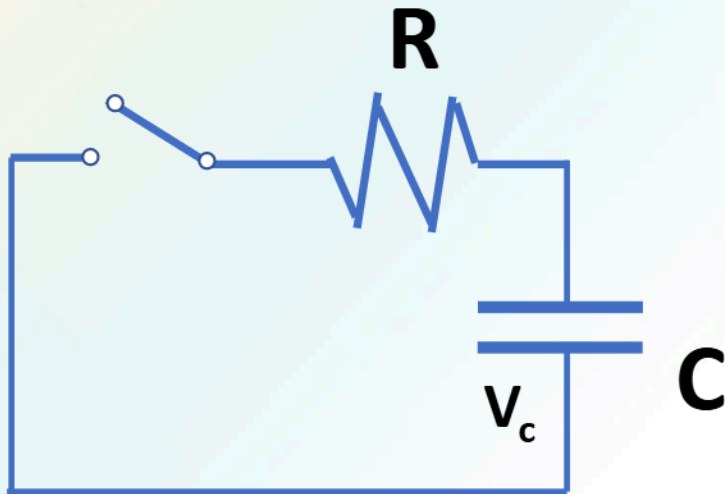


The net voltage available to charge the capacitor is the difference between the capacitor voltage V_c and V . As V_c , capacitor's emf increases. Consequently, the rate of charge slows down. This cumulative effect continues until V_c is approximately equal to V , at which time the charging current I_c is reduced to nearly zero.

$$V_c = V(1 - e^{-\frac{t}{\tau}})$$

Discharging Capacitors

When a capacitor is disconnected from the source V , and the switch is on, the capacitor is discharging.



The Voltage between the Plates drops down gradually to zero, $V_c = 0$.

$$V_c = V e^{-\frac{t}{\tau}}$$



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Data and Results

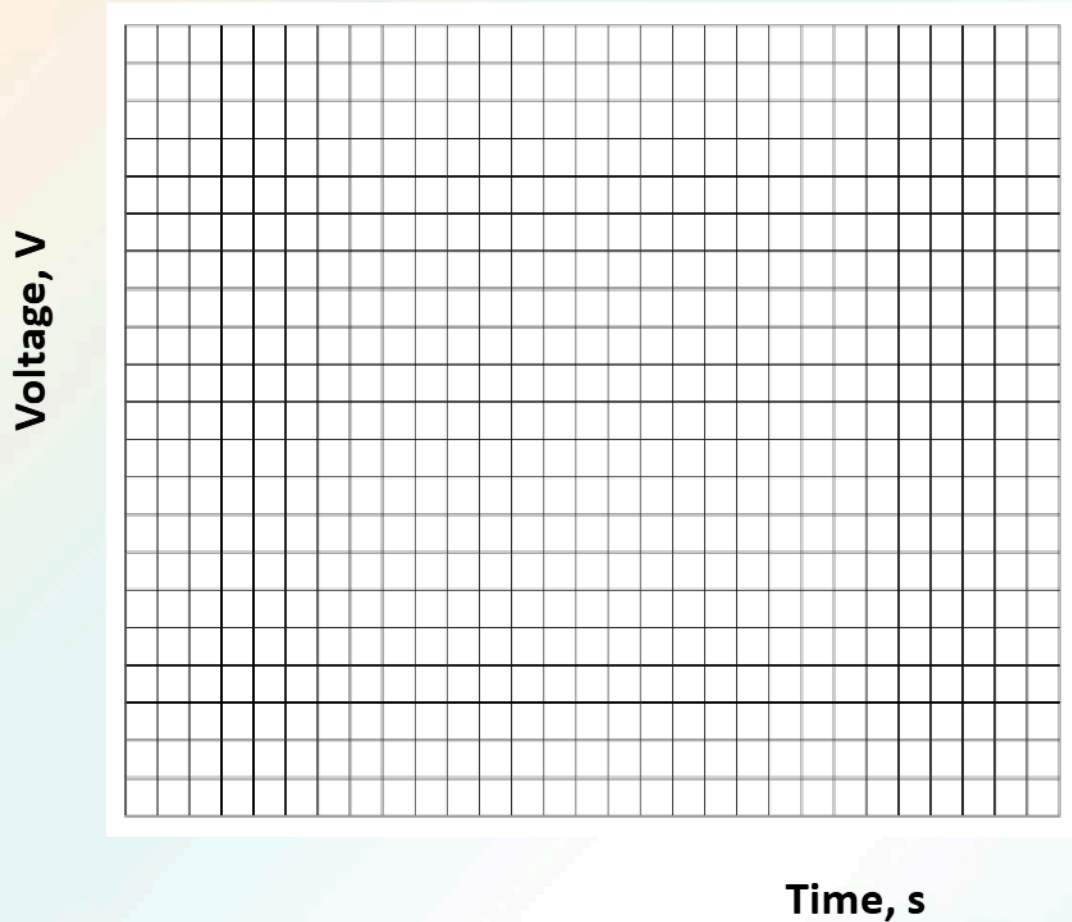
Use the following table and templates for the data and results

Time (s)			Voltage, V	
	sec	min:sec	Charging	Discharging
0.25 τ	55	0:55		
0.50 τ	110	1:50		
0.75 τ	165	2:45		
1.00 τ	220	3:40		
1.50 τ	330	5:30		
2.00 τ	440	7:20		
2.50 τ	550	9:10		
3.00 τ	660	11:00		
3.50 τ	770	12:50		
4.00 τ	880	14:40		
4.50 τ	990	16:30		
5.00 τ	1100	18:10		

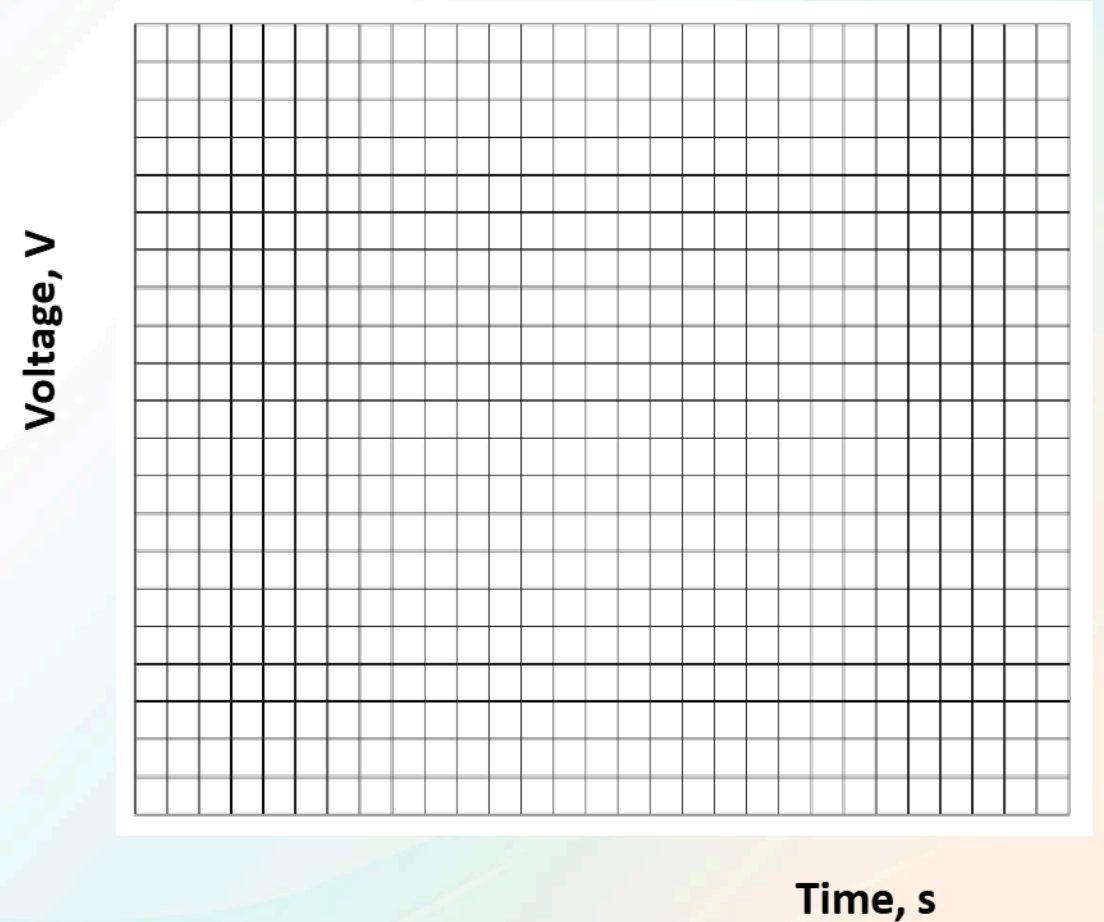


Data and Results

Charging Voltage of a Capacitor versus Time



Discharging Voltage of a Capacitor versus Time



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REFERENCES

Electricalacademia, *RC Circuit Time Constant/ Charging and Discharging of Capacitor*. Retrieved from <https://electricalacademia.com/basic-electrical/rc-circuit-time-constant-rc-time-constant/>

Sophie, *Charging and Discharging Capacitors*. Science and Math Revision. Retrieved from <https://www.scienceandmathsrevision.co.uk/topic/charging-and-discharging-capacitors/>



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